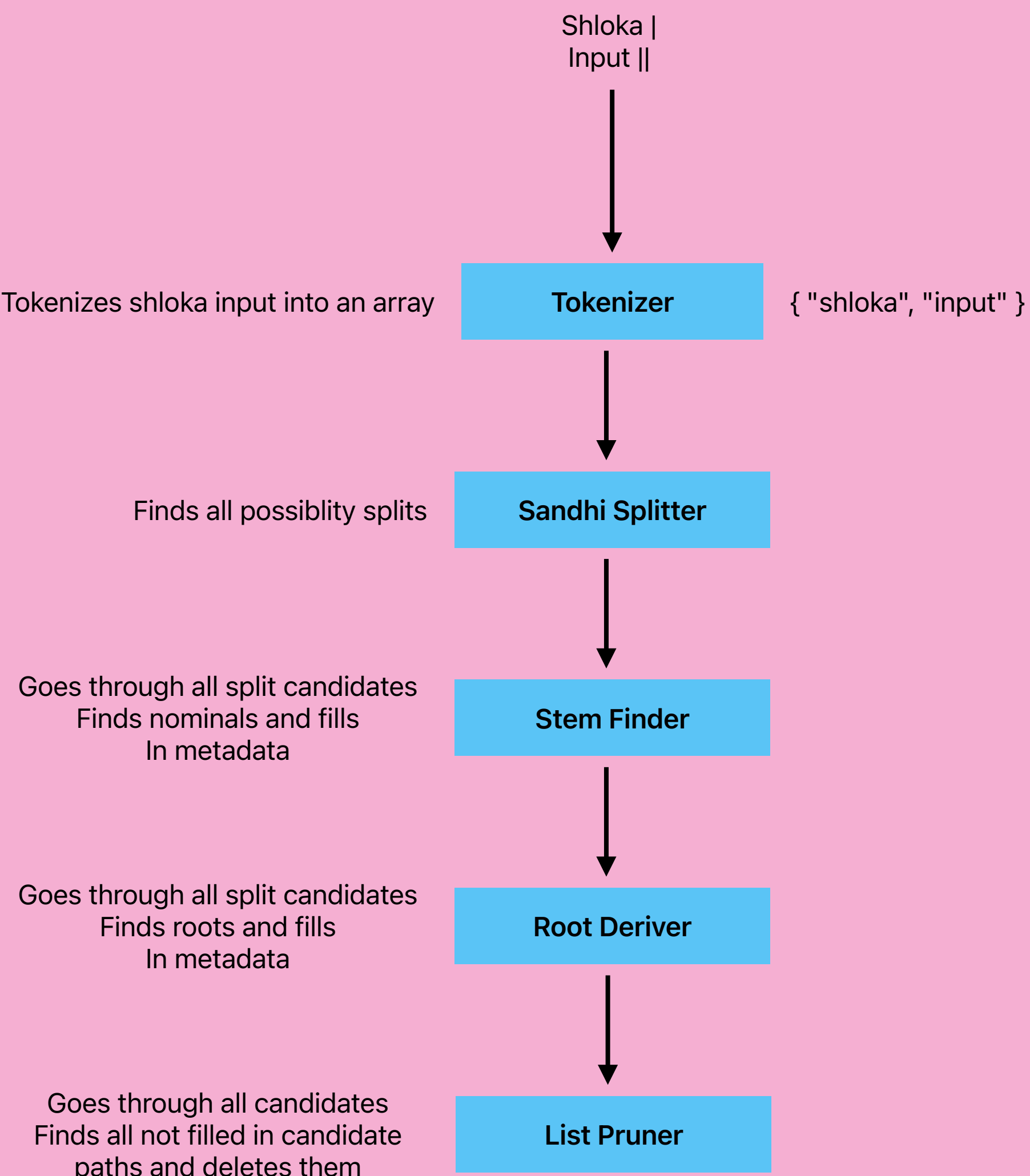
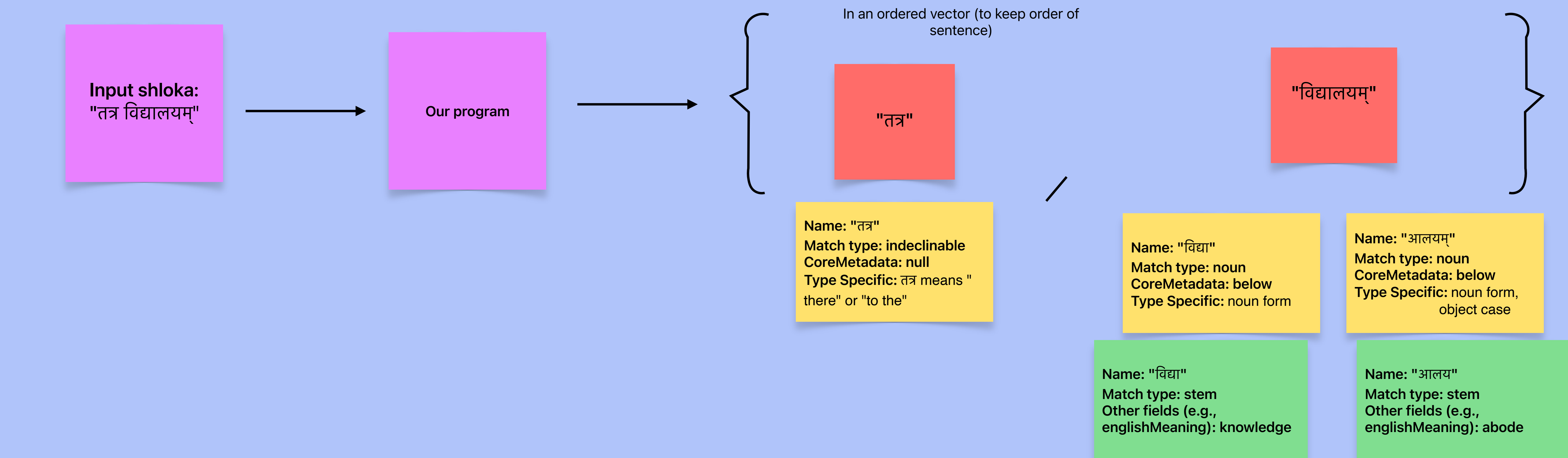


Pipeline



Data Structures



**Word Type encompass**

**Who:**  
The word

**What:**  
Contains all the information about the word

**When:**  
Throughout the program, we can use this to describe all our findings of the word with one data structure. Full unsandhi'd word (it will contain a field that has all sandhi'd possibilities).

**Where:**  
The program

**Why:**  
So that when we try to calculate the meaning of a word we have just one structure that contains all our word and keep all of that in order

**Word Type Fields**

- Word name
- vector<vector<Metadatas>>
  - so that sandhi splitter can put all of its possibilities (e.g., { { "Hita", "Upadesha" }, { "Hitopa", "adesha" } }

**Word Metadata**

**Who:**  
One specific unsandhi'd form word

**What:**  
Contains one path that can derive the word

**When/Where:**  
Mainly Root Deriver/  
Sandhi Splitter

**Why:**  
One specific way that the user can understand the word

**Word Metadata Fields**

- Name
- Match type (noun, verb)
- CoreMetadata
  - Only 1 because there can only be one root/stem
- Type specific metadata
  - Tense for verbs, case for nouns
    - Determined on suffix/prefix

**Core Metadata**

**Who:**  
The root/stem that makes up a word

**What:**  
Contains all the information about

**When/Where:**  
Mainly Root Deriver/  
Stem Splitter

**Why:**  
So that the user knows where the word came from

**Core Metadata Fields**

- Name
- Match type (stem, root)
- All fields in "roots.json" or "stems.json"